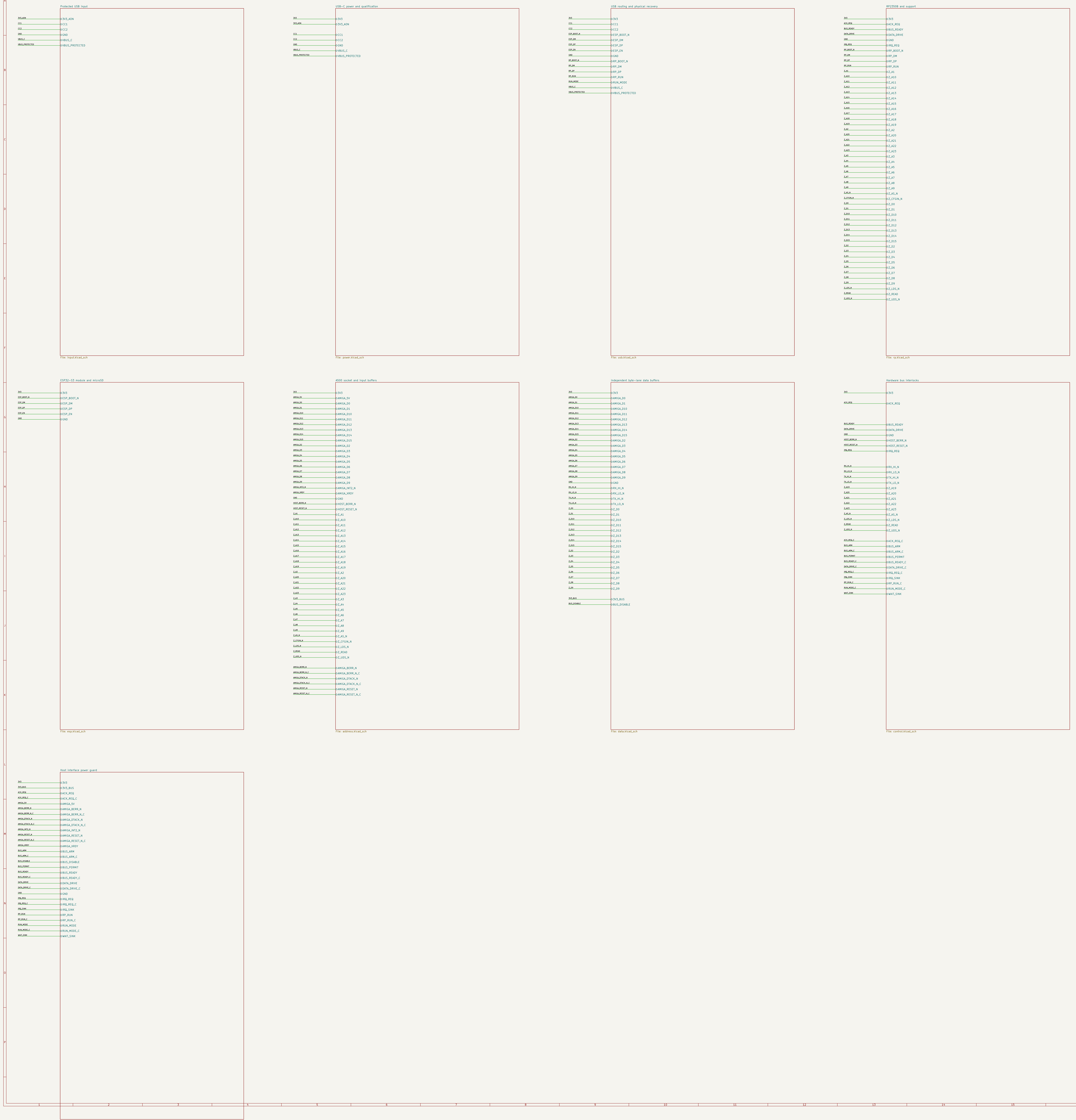
FujiNet A500 integrated R4 — circuit review; NOT fabrication release

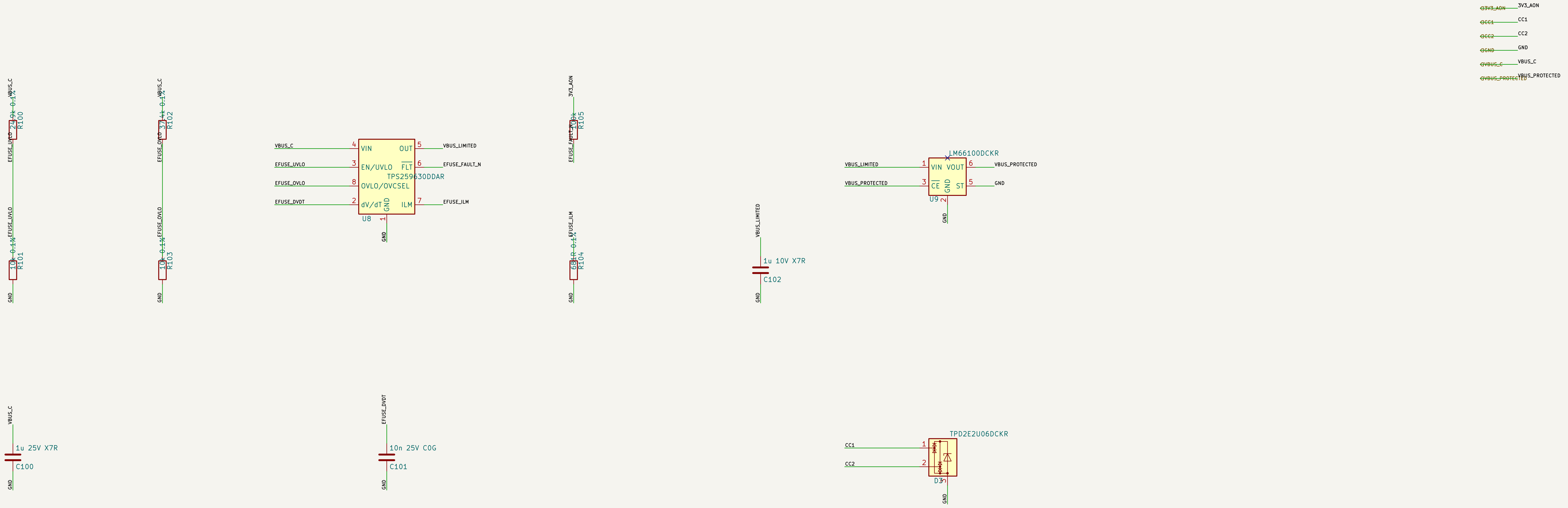

Protected USB input
3V3_AON
CC1
CC2
GND
VBUS_C
VBUS_PROTECTED
File: input.kicad_sch
USB−C power and qualification
3V3
3V3_AON
CC1
CC2
GND
VBUS_C
VBUS_PROTECTED
File: power.kicad_sch
USB routing and physical recovery
3V3
CC1
CC2
ESP_BOOT_N
ESP_DM
ESP_DP
ESP_EN
GND
RP_BOOT_N
RP_DM
RP_DP
RP_RUN
RUN_MODE
VBUS_C
VBUS_PROTECTED
File: usb.kicad_sch
RP2350B and support
3V3
ACK_REQ
BUS_READY
DATA_DRIVE
GND
IRQ_REQ
RP_BOOT_N
RP_DM
RP_DP
RP_RUN
Z_A1
Z_A10
Z_A11
Z_A12
Z_A13
Z_A14
Z_A15
Z_A16
Z_A17
Z_A18
Z_A19
Z_A2
Z_A20
Z_A21
Z_A22
Z_A23
Z_A3
Z_A4
Z_A5
Z_A6
Z_A7
Z_A8
Z_A9
Z_AS_N
Z_CFGIN_N
Z_D0
Z_D1
Z_D10
Z_D11
Z_D12
Z_D13
Z_D14
Z_D15
Z_D2
Z_D3
Z_D4
Z_D5
Z_D6
Z_D7
Z_D8
Z_D9
Z_LDS_N
Z_READ
Z_UDS_N
File: rp.kicad_sch
ESP32−S3 module and microSD
3V3
ESP_BOOT_N
ESP_DM
ESP_DP
ESP_EN
GND
File: esp.kicad_sch
A500 socket and input buffers
3V3
AMIGA_5V
AMIGA_D0
AMIGA_D1
AMIGA_D10
AMIGA_D11
AMIGA_D12
AMIGA_D13
AMIGA_D14
AMIGA_D15
AMIGA_D2
AMIGA_D3
AMIGA_D4
AMIGA_D5
AMIGA_D6
AMIGA_D7
AMIGA_D8
AMIGA_D9
AMIGA_INT2_N
AMIGA_XRDY
GND
HOST_BERR_N
HOST_RESET_N
Z_A1
Z_A10
Z_A11
Z_A12
Z_A13
Z_A14
Z_A15
Z_A16
Z_A17
Z_A18
Z_A19
Z_A2
Z_A20
Z_A21
Z_A22
Z_A23
Z_A3
Z_A4
Z_A5
Z_A6
Z_A7
Z_A8
Z_A9
Z_AS_N
Z_CFGIN_N
Z_LDS_N
Z_READ
Z_UDS_N
AMIGA_BERR_N
AMIGA_BERR_N_C
AMIGA_DTACK_N
AMIGA_DTACK_N_C
AMIGA_RESET_N
AMIGA_RESET_N_C
File: address.kicad_sch
Independent byte−lane data buffers
3V3
AMIGA_D0
AMIGA_D1
AMIGA_D10
AMIGA_D11
AMIGA_D12
AMIGA_D13
AMIGA_D14
AMIGA_D15
AMIGA_D2
AMIGA_D3
AMIGA_D4
AMIGA_D5
AMIGA_D6
AMIGA_D7
AMIGA_D8
AMIGA_D9
GND
RX_HI_N
RX_LO_N
TX_HI_N
TX_LO_N
Z_D0
Z_D1
Z_D10
Z_D11
Z_D12
Z_D13
Z_D14
Z_D15
Z_D2
Z_D3
Z_D4
Z_D5
Z_D6
Z_D7
Z_D8
Z_D9
3V3_BUS
BUS_DISABLE
File: data.kicad_sch
Hardware bus interlocks
3V3
ACK_REQ
BUS_READY
DATA_DRIVE
GND
HOST_BERR_N
HOST_RESET_N
IRQ_REQ
RX_HI_N
RX_LO_N
TX_HI_N
TX_LO_N
Z_A19
Z_A20
Z_A21
Z_A22
Z_A23
Z_AS_N
Z_LDS_N
Z_READ
Z_UDS_N
ACK_REQ_C
BUS_ARM
BUS_ARM_C
BUS_PERMIT
BUS_READY_C
DATA_DRIVE_C
IRQ_REQ_C
IRQ_SINK
RP_RUN_C
RUN_MODE_C
WAIT_SINK
File: control.kicad_sch
Host interface power guard
3V3
3V3_BUS
ACK_REQ
ACK_REQ_C
AMIGA_5V
AMIGA_BERR_N
AMIGA_BERR_N_C
AMIGA_DTACK_N
AMIGA_DTACK_N_C
AMIGA_INT2_N
AMIGA_RESET_N
AMIGA_RESET_N_C
AMIGA_XRDY
BUS_ARM
BUS_ARM_C
BUS_DISABLE
BUS_PERMIT
BUS_READY
BUS_READY_C
DATA_DRIVE
DATA_DRIVE_C
GND
IRQ_REQ
IRQ_REQ_C
IRQ_SINK
RP_RUN
RP_RUN_C
RUN_MODE
RUN_MODE_C
WAIT_SINK

Sheet: /
File: integrated−a500−r4.kicad_sch

Title: FujiNet direct A500 integrated candidate

Size: A0	Date: 2026−11−09	**Rev: R4 candidate**
KiCad E.D.A. 10.0.7		Id: 1/10

USB VBUS -> current/voltage protection -> reverse-blocking ideal diode -> AON and qualified main supply.

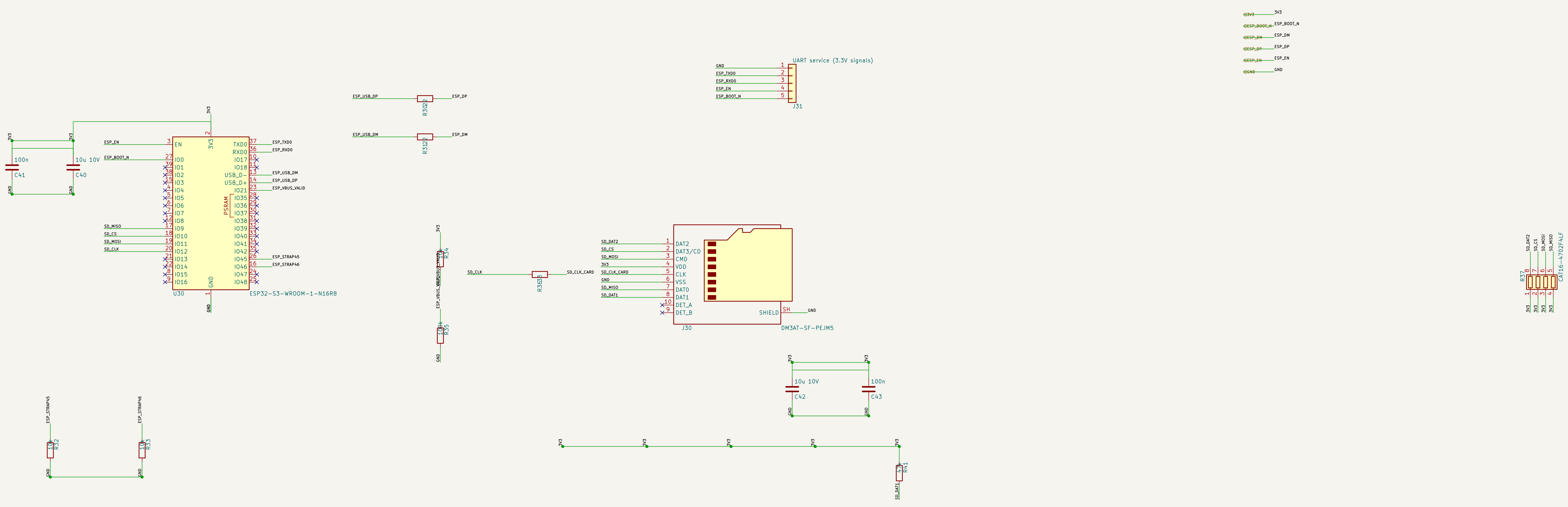

3V3_AON
CC1
CC2
GND
VBUS_C
VBUS_PROTECTED
VBUS_C
24.9k 0.1%
R100
EFUSE_UVLO
10k 0.1%
R101
GND
VBUS_C
37.4k 0.1%
R102
EFUSE_OVLO
10k 0.1%
R103
GND
VBUS_C
EFUSE_UVLO
EFUSE_OVLO
EFUSE_DVDT
VIN
EN/UVLO
OVLO/OVCSEL
dV/dT
GND
OUT
FLT
ILM
TPS259630DDAR
U8
VBUS_LIMITED
EFUSE_FAULT_N
EFUSE_ILM
GND
3V3_AON
100k
R105
EFUSE_FAULT_N
EFUSE_ILM
681R 0.1%
R104
GND
VBUS_LIMITED
1u 10V X7R
C102
GND
LM66100DCKR
VBUS_LIMITED
VBUS_PROTECTED
VIN
CE
GND
VOUT
ST
U9
VBUS_PROTECTED
GND
GND
1u 25V X7R
C100
GND
EFUSE_DVDT
10n 25V C0G
C101
GND
TPD2E2U06DCKR
CC1
CC2
D3
GND

681R: nominal 1.337A limit. OVLO nominal 5.688V. Latch–off thermal fault: unplug USB to recover.

No Amiga supply connection. E–fuse pad/package and fast overvoltage transient review required before layout.

FujiNet NIO
Sheet: /Protected USB input/
File: input.kicad_sch
Title: Protected USB input
Size: A2 Date: 2026–11–09 **Rev: R4 candidate**
KiCad E.D.A. 10.0.7 Id: 2/10

N16R8 module includes flash, PSRAM, crystal and RF. IO35/36/37 are reserved for octal PSRAM.


3V3
GND
100n
C41
10u 10V
C40
ESP_EN
ESP_BOOT_N
SD_MISO
SD_CS
SD_MOSI
SD_CLK
EN
3V3
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
TXD0
RXD0
IO17
IO18
USB_D-
USB_D+
IO21
PSRAM
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
IO45
IO46
IO47
IO48
GND
U30
ESP32-S3-WROOM-1-N16R8
ESP_TXD0
ESP_RXD0
ESP_USB_DM
ESP_USB_DP
ESP_VBUS_VALID
ESP_STRAP45
ESP_STRAP46
ESP_USB_DP
R30 22
ESP_DP
ESP_USB_DM
R31 22
ESP_DM
10k
R34
ESP_VBUS_VALID
100k
R35
SD_CLK
R36 33
SD_CLK_CARD
UART service (3.3V signals)
GND
ESP_TXD0
ESP_RXD0
ESP_EN
ESP_BOOT_N
J31
SD_DAT2
SD_CS
SD_MOSI
3V3
SD_CLK_CARD
GND
SD_MISO
SD_DAT1
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
SHIELD
SH
J30
DM3AT-SF-PEJM5
10u 10V
C42
100n
C43
ESP_STRAP45
ESP_STRAP46
10k
R32
10k
R33
47k
R41
SD_DAT1
3V3
ESP_BOOT_N
ESP_DM
ESP_DP
ESP_EN
GND
R37
CAT16-4702F4LF

Antenna at enclosure/PCB edge: apply Espressif all-layer antenna keepout before PCB placement.

FujiNet NIO
Sheet: /ESP32-S3 module and microSD/
File: esp.kicad_sch
Title: ESP32-S3 module and microSD
Size: A1 | Date: 2026-11-09 | **Rev: R4 candidate**
KiCad E.D.A. 10.0.7 | Id: 3/10

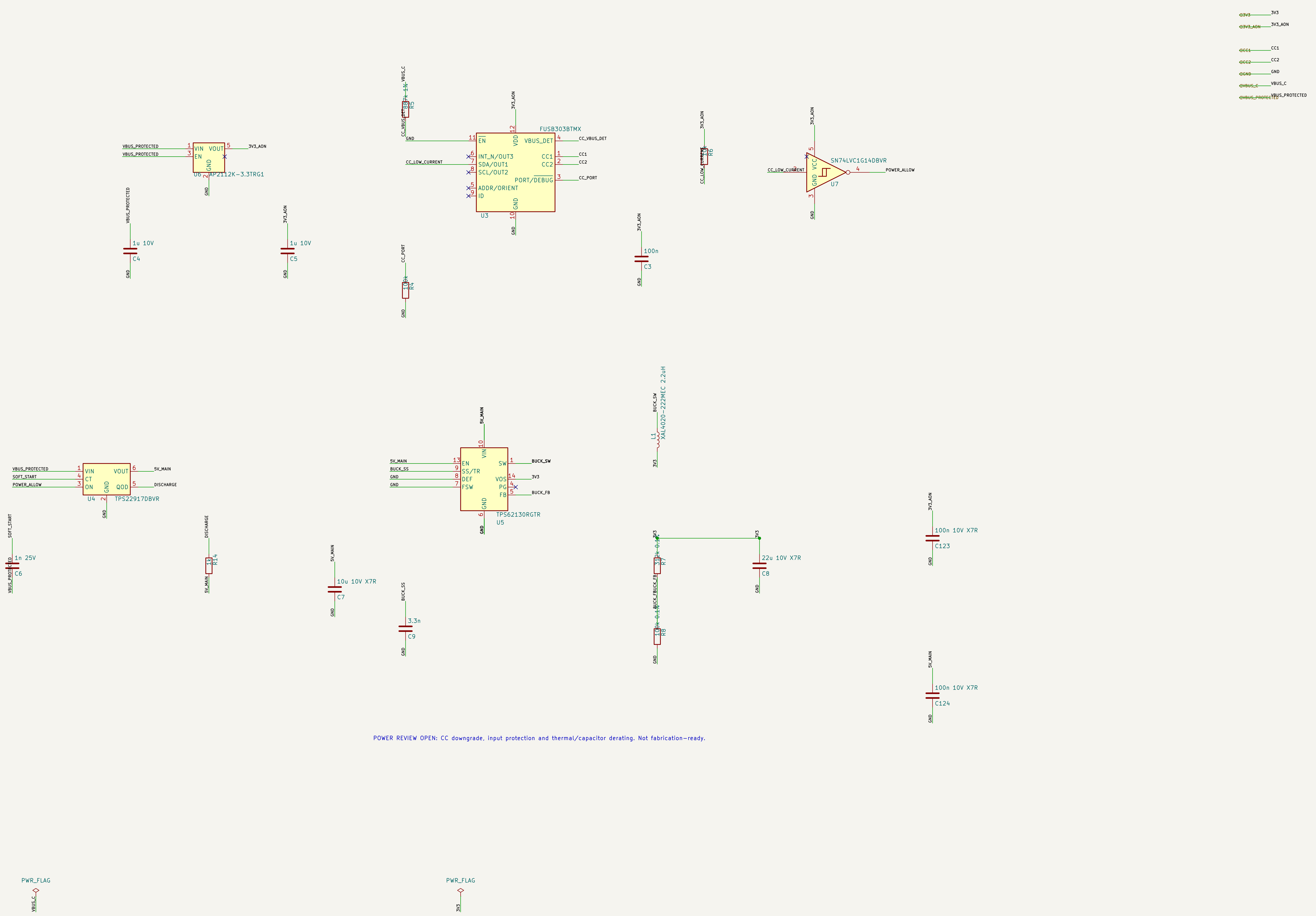
TPS25530DBVR
U95
AMIGA_5V
HOST_ILIM
HOST_5V_LIMITED
100n 10V X7R
C110
SS14-13-F
D4
AP2112K-3.3TRG1
U96
3V3_BUS
1u 10V X7R
C111
1u 10V X7R
C112
PWR_FLAG
100u 6.3V polymer 20% ESR<=50m
C113
MIC2790H-04VD6-TR
U60
GUARD_OK_RAW
100p 25V C0G 5%
C114
1u 10V X7R
C60
MIC2790H-04VD6-TR
U90
100p 25V C0G 5%
C115
1u 10V X7R
C118
MIC2790H-04VD6-TR
U91
100p 25V C0G 5%
C116
1u 10V X7R
C119
SN74LVC2G14DBVR
U92
BUS_PERMIT
BUS_DISABLE
100n 10V X7R
C120
SN74LVC2G132DCUR
U93
IRQ_OE_N
WAIT_OE_N
100n 10V X7R
C121
SN74LVC2G125DCUR
U94
AMIGA_INT2_N
AMIGA_XRDY
100n 10V X7R
C122
MMBT3904-7-F
Q5
220u 6.3V polymer 20% ESR<=50m
C117
74LVC3G17DC,125
U97
BUS_ARM_C
RUN_MODE_C
RP_RUN_C
100n 10V X7R
C125
74LVC3G17DC,125
U98
ACK_REQ_C
DATA_DRIVE_C
BUS_READY_C
100n 10V X7R
C126
SN74LVC1G17DBVR
U99
NC
IRQ_REQ
IRQ_REQ_C
100n 10V X7R
C127
74LVC3G17DC,125
U100
AMIGA_DTACK_N_C
AMIGA_BERR_N_C
AMIGA_RESET_N_C
100n 10V X7R
C128
3V3
3V3_BUS
ACK_REQ
ACK_REQ_C
AMIGA_5V
AMIGA_BERR_N
AMIGA_BERR_N_C
AMIGA_DTACK_N
AMIGA_DTACK_N_C
AMIGA_INT2_N
AMIGA_RESET_N
AMIGA_RESET_N_C
AMIGA_XRDY
BUS_ARM
BUS_ARM_C
BUS_DISABLE
BUS_PERMIT
BUS_READY
BUS_READY_C
DATA_DRIVE
DATA_DRIVE_C
GND
IRQ_REQ
IRQ_REQ_C
IRQ_SINK
RP_RUN
RP_RUN_C
RUN_MODE
RUN_MODE_C
WAIT_SINK

Sheet: /Host interface power guard/
File: guard.kicad_sch

Title: guard

Size: A1 | Date: | **Rev:**
KiCad E.D.A. 10.0.7 | Id: 4/10

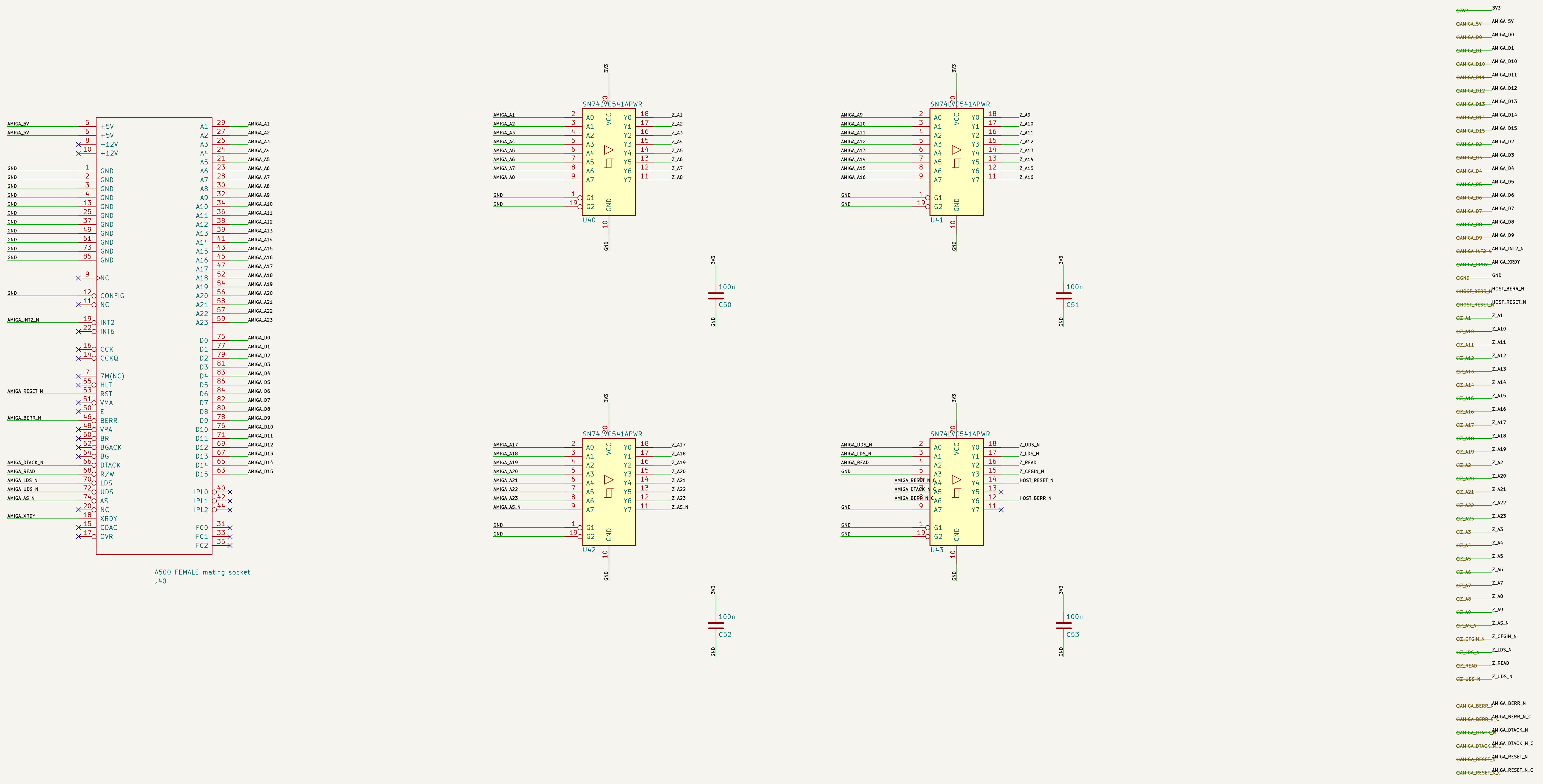
USB−C sink only. >=1.5 A advertised required. No Amiga supply connection to these rails.


VBUS_PROTECTED
VIN
EN
VOUT
GND
3V3_AON
U6
AP2112K−3.3TRG1
1u 10V
C4
C5
VBUS_C
887k 1%
R5
CC_VBUS_DET
FUSB303BTMX
U3
VDD
INT_N/OUT3
SDA/OUT1
SCL/OUT2
ADDR/ORIENT
ID
VBUS_DET
CC1
CC2
PORT/DEBUG
CC_LOW_CURRENT
CC_PORT
100k
R4
100n
C3
10k
R6
SN74LVC1G14DBVR
U7
VCC
POWER_ALLOW
3V3
CC1
CC2
GND
VBUS_C
VBUS_PROTECTED
TPS22917DBVR
U4
CT
ON
QOD
SOFT_START
DISCHARGE
5V_MAIN
1n 25V
C6
1k
R14
10u 10V X7R
C7
TPS62130RGTR
U5
SS/TR
DEF
FSW
SW
VOS
PG
FB
BUCK_SS
BUCK_SW
BUCK_FB
3.3n
C9
L1
XAL4020−222MEC 2.2uH
312k 0.1%
R7
100k 0.1%
R8
22u 10V X7R
C8
100n 10V X7R
C123
C124
PWR_FLAG
PWR_FLAG

POWER REVIEW OPEN: CC downgrade, input protection and thermal/capacitor derating. Not fabrication−ready.

FujiNet NIO
Sheet: /USB−C power and qualification/
File: power.kicad_sch
Title: USB−C power and qualification
Size: A1 | Date: 2026−11−09 | **Rev: R4 candidate**
KiCad E.D.A. 10.0.7 | Id: 5/10

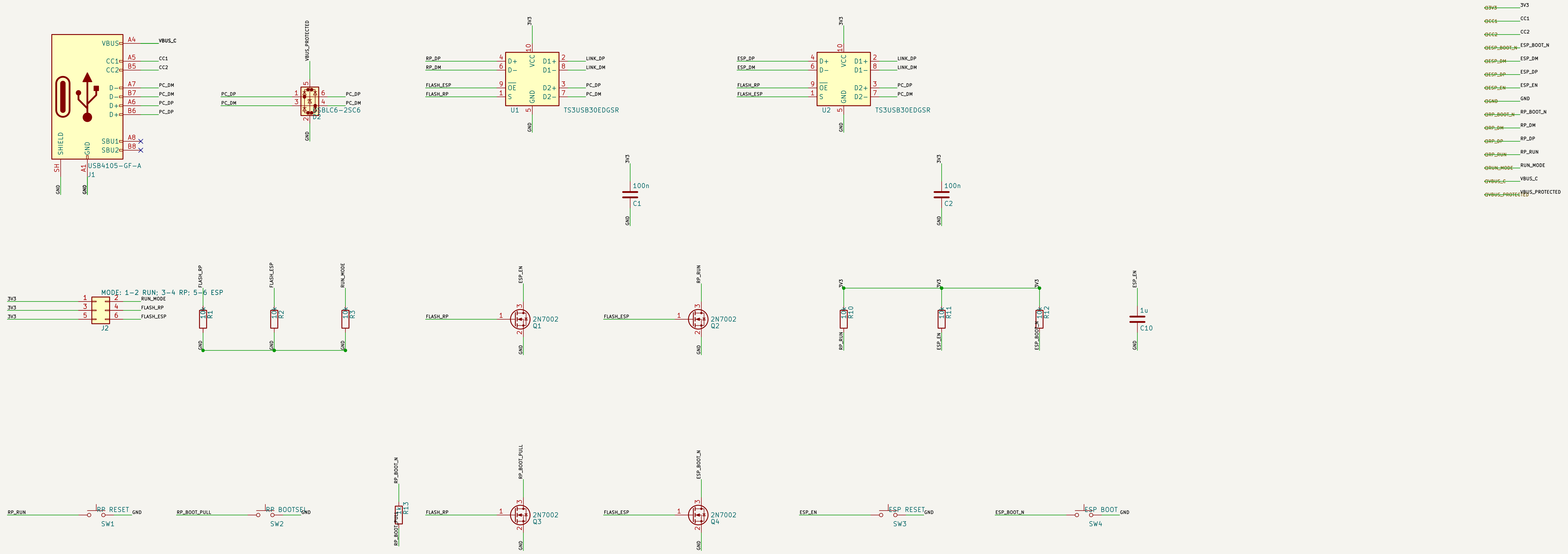
86–contact A500 socket; NOT the existing male PCB–edge footprint. No A2000 SLAVE/CFGOUT here.


A500 FEMALE mating socket
J40
SN74LVC541APWR
U40
U41
U42
U43
3V3
GND
100n
C50
C51
C52
C53

U40–U43 use pin–compatible 541 logic symbol; fitted part MUST be 3.3V SN74LVC541A (5V tolerant, Ioff).

FujiNet NIO
Sheet: /A500 socket and input buffers/
File: address.kicad_sch
Title: A500 socket and input buffers
Size: A1
Date: 2026–11–09
Rev: R4 candidate
KiCad E.D.A. 10.0.7
Id: 6/10

Shared USB: cold mode changes only. Unplug power before moving the single MODE shunt.


VBUS
CC1
CC2
D−
D−
D+
D+
SBU1
SBU2
SHIELD
GND
A4
A5
B5
A7
B7
A6
B6
A8
B8
SH
A1
VBUS_C
CC1
CC2
PC_DM
PC_DM
PC_DP
PC_DP
GND
GND
USB4105-GF-A
J1
VBUS_PROTECTED
PC_DP
PC_DM
PC_DP
PC_DM
USBLC6-2SC6
D2
GND
3V3
RP_DP
RP_DM
FLASH_ESP
FLASH_RP
D+
D−
OE
S
VCC
GND
D1+
D1−
D2+
D2−
LINK_DP
LINK_DM
PC_DP
PC_DM
U1
TS3USB30EDGSR
GND
3V3
ESP_DP
ESP_DM
FLASH_RP
FLASH_ESP
LINK_DP
LINK_DM
PC_DP
PC_DM
U2
TS3USB30EDGSR
GND
3V3
100n
C1
GND
3V3
100n
C2
GND
3V3
CC1
CC2
ESP_BOOT_N
ESP_DM
ESP_DP
ESP_EN
GND
RP_BOOT_N
RP_DM
RP_DP
RP_RUN
RUN_MODE
VBUS_C
VBUS_PROTECTED
MODE: 1–2 RUN; 3–4 RP; 5–6 ESP
3V3
3V3
3V3
RUN_MODE
FLASH_RP
FLASH_ESP
J2
FLASH_RP
FLASH_ESP
RUN_MODE
10k
R1
10k
R2
10k
R3
GND
GND
GND
ESP_EN
FLASH_RP
2N7002
Q1
GND
RP_RUN
FLASH_ESP
2N7002
Q2
GND
3V3
3V3
3V3
10k
R10
10k
R11
10k
R12
RP_RUN
ESP_EN
ESP_BOOT_N
ESP_EN
1u
C10
GND
RP_RUN
RP RESET
SW1
GND
RP_BOOT_PULL
RP BOOTSEL
SW2
GND
RP_BOOT_N
1k
R13
RP_BOOT_PULL
RP_BOOT_PULL
FLASH_RP
2N7002
Q3
GND
ESP_BOOT_N
FLASH_ESP
2N7002
Q4
GND
ESP_EN
ESP RESET
SW3
GND
ESP_BOOT_N
ESP BOOT
SW4
GND

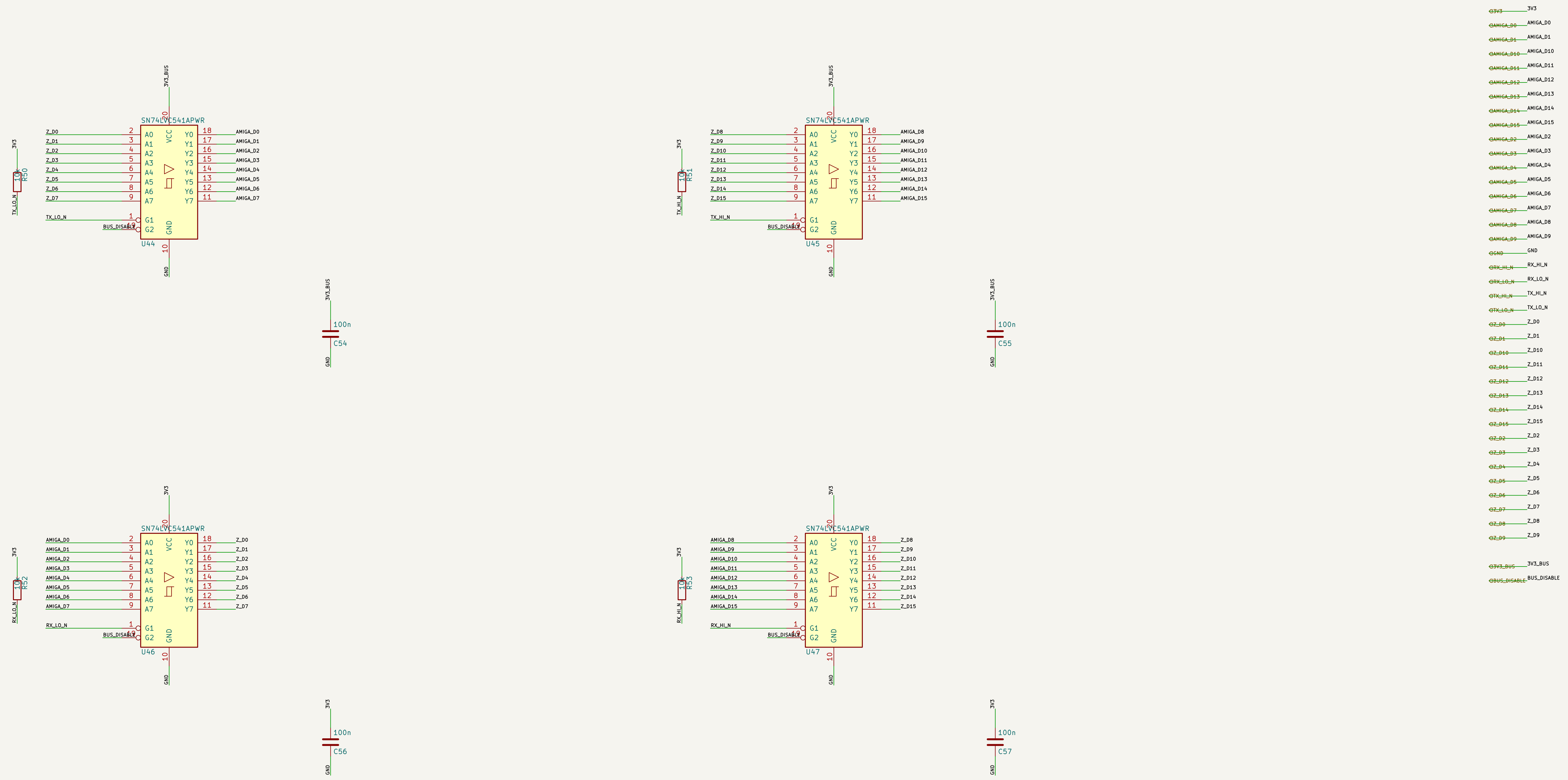
Physical recovery: selected BOOT held low, peer held RESET. SW1/SW3 are independent resets.

FujiNet NIO
Sheet: /USB routing and physical recovery/
File: usb.kicad_sch

Title: USB routing and physical recovery

Size: A2	Date: 2026-11-09	**Rev: R4 candidate**
KiCad E.D.A. 10.0.7		Id: 7/10

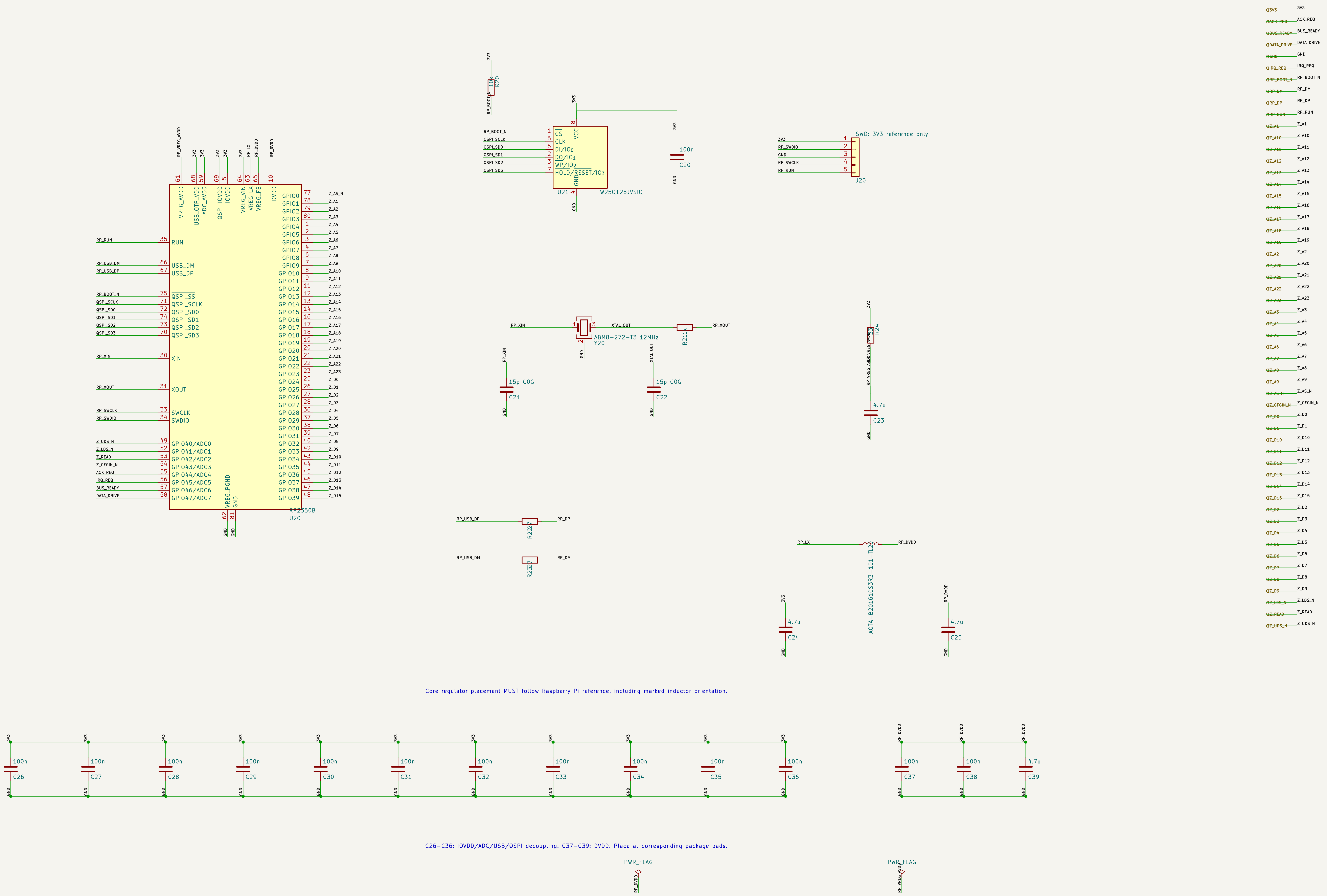
Upper row: RP -> Amiga. Lower row: Amiga -> RP. Byte enables independent; hardware read/write qualification.


3V3_BUS
SN74LVC541APWR
U44
Z_D0
Z_D1
Z_D2
Z_D3
Z_D4
Z_D5
Z_D6
Z_D7
TX_LO_N
BUS_DISABLE
A0
A1
A2
A3
A4
A5
A6
A7
G1
G2
VCC
GND
Y0
Y1
Y2
Y3
Y4
Y5
Y6
Y7
2
3
4
5
6
7
8
9
1
19
20
10
18
17
16
15
14
13
12
11
AMIGA_D0
AMIGA_D1
AMIGA_D2
AMIGA_D3
AMIGA_D4
AMIGA_D5
AMIGA_D6
AMIGA_D7
3V3
10k
R50
TX_LO_N
GND
3V3_BUS
100n
C54
GND
SN74LVC541APWR
U45
Z_D8
Z_D9
Z_D10
Z_D11
Z_D12
Z_D13
Z_D14
Z_D15
TX_HI_N
BUS_DISABLE
AMIGA_D8
AMIGA_D9
AMIGA_D10
AMIGA_D11
AMIGA_D12
AMIGA_D13
AMIGA_D14
AMIGA_D15
3V3
10k
R51
TX_HI_N
100n
C55
SN74LVC541APWR
U46
AMIGA_D0
AMIGA_D1
AMIGA_D2
AMIGA_D3
AMIGA_D4
AMIGA_D5
AMIGA_D6
AMIGA_D7
RX_LO_N
BUS_DISABLE
Z_D0
Z_D1
Z_D2
Z_D3
Z_D4
Z_D5
Z_D6
Z_D7
3V3
10k
R52
RX_LO_N
100n
C56
SN74LVC541APWR
U47
AMIGA_D8
AMIGA_D9
AMIGA_D10
AMIGA_D11
AMIGA_D12
AMIGA_D13
AMIGA_D14
AMIGA_D15
RX_HI_N
BUS_DISABLE
Z_D8
Z_D9
Z_D10
Z_D11
Z_D12
Z_D13
Z_D14
Z_D15
3V3
10k
R53
RX_HI_N
100n
C57
3V3
AMIGA_D0
AMIGA_D1
AMIGA_D10
AMIGA_D11
AMIGA_D12
AMIGA_D13
AMIGA_D14
AMIGA_D15
AMIGA_D2
AMIGA_D3
AMIGA_D4
AMIGA_D5
AMIGA_D6
AMIGA_D7
AMIGA_D8
AMIGA_D9
GND
RX_HI_N
RX_LO_N
TX_HI_N
TX_LO_N
Z_D0
Z_D1
Z_D10
Z_D11
Z_D12
Z_D13
Z_D14
Z_D15
Z_D2
Z_D3
Z_D4
Z_D5
Z_D6
Z_D7
Z_D8
Z_D9
3V3_BUS
BUS_DISABLE

PIO: preload read data/direction, pulse DATA_DRIVE, settle, pulse ACK_REQ. Return GPIO to input between cycles.

FujiNet NIO
Sheet: /Independent byte–lane data buffers/
File: data.kicad_sch
Title: Independent byte–lane data buffers
Size: A1
Date: 2026–11–09
Rev: R4 candidate
KiCad E.D.A. 10.0.7
Id: 8/10

48 GPIO: address/data/control only; USB and SWD dedicated pads. GPIO45 IRQ; GPIO46 BUS_READY.


RP2350B
U20
RP_VREG_AVDD 61 VREG_AVDD
3V3 68 USB_OTP_VDD
3V3 59 ADC_AVDD
3V3 69 QSPI_IOVDD
3V3 5 IOVDD
3V3 64 VREG_VIN
RP_LX 63 VREG_LX
RP_DVDD 65 VREG_FB
RP_DVDD 10 DVDD
RP_RUN 35 RUN
RP_USB_DM 66 USB_DM
RP_USB_DP 67 USB_DP
RP_BOOT_N 75 QSPI_SS
QSPI_SCLK 71 QSPI_SCLK
QSPI_SD0 72 QSPI_SD0
QSPI_SD1 74 QSPI_SD1
QSPI_SD2 73 QSPI_SD2
QSPI_SD3 70 QSPI_SD3
RP_XIN 30 XIN
RP_XOUT 31 XOUT
RP_SWCLK 33 SWCLK
RP_SWDIO 34 SWDIO
Z_UDS_N 49 GPIO40/ADC0
Z_LDS_N 52 GPIO41/ADC1
Z_READ 53 GPIO42/ADC2
Z_CFGIN_N 54 GPIO43/ADC3
ACK_REQ 55 GPIO44/ADC4
IRQ_REQ 56 GPIO45/ADC5
BUS_READY 57 GPIO46/ADC6
DATA_DRIVE 58 GPIO47/ADC7
GPIO0 77 Z_AS_N
GPIO1 78 Z_A1
GPIO2 79 Z_A2
GPIO3 80 Z_A3
GPIO4 1 Z_A4
GPIO5 2 Z_A5
GPIO6 3 Z_A6
GPIO7 4 Z_A7
GPIO8 6 Z_A8
GPIO9 7 Z_A9
GPIO10 8 Z_A10
GPIO11 9 Z_A11
GPIO12 11 Z_A12
GPIO13 12 Z_A13
GPIO14 13 Z_A14
GPIO15 14 Z_A15
GPIO16 16 Z_A16
GPIO17 17 Z_A17
GPIO18 18 Z_A18
GPIO19 19 Z_A19
GPIO20 20 Z_A20
GPIO21 21 Z_A21
GPIO22 22 Z_A22
GPIO23 23 Z_A23
GPIO24 25 Z_D0
GPIO25 26 Z_D1
GPIO26 27 Z_D2
GPIO27 28 Z_D3
GPIO28 36 Z_D4
GPIO29 37 Z_D5
GPIO30 38 Z_D6
GPIO31 39 Z_D7
GPIO32 40 Z_D8
GPIO33 42 Z_D9
GPIO34 43 Z_D10
GPIO35 44 Z_D11
GPIO36 45 Z_D12
GPIO37 46 Z_D13
GPIO38 47 Z_D14
GPIO39 48 Z_D15
VREG_PGND 62 GND
GND 81 GND
3V3
10k
R20
RP_BOOT_N
3V3
RP_BOOT_N 1 CS
QSPI_SCLK 6 CLK
QSPI_SD0 5 DI/IO0
QSPI_SD1 2 DO/IO1
QSPI_SD2 3 WP/IO2
QSPI_SD3 7 HOLD/RESET/IO3
8 VCC
4 GND
GND
U21
W25Q128JVSIQ
3V3
100n
C20
GND
SWD: 3V3 reference only
3V3 1
RP_SWDIO 2
GND 3
RP_SWCLK 4
RP_RUN 5
J20
RP_XIN
1
3
2
XTAL_OUT
ABM8-272-T3 12MHz
Y20
GND
R21
1k
RP_XOUT
RP_XIN
15p C0G
C21
GND
XTAL_OUT
15p C0G
C22
GND
3V3
R24
33
RP_VREG_AVDD
4.7u
C23
GND
RP_USB_DP
R22
27
RP_DP
RP_USB_DM
R23
27
RP_DM
RP_LX
L20
AOTA-B201610S3R3-101-T
RP_DVDD
3V3
4.7u
C24
GND
RP_DVDD
4.7u
C25
GND
3V3
100n
C26
C27
C28
C29
C30
C31
C32
C33
C34
C35
C36
GND
RP_DVDD
100n
C37
100n
C38
4.7u
C39
GND
PWR_FLAG
RP_DVDD
PWR_FLAG
RP_VREG_AVDD
~~3V3~~ 3V3
~~ACK_REQ~~ ACK_REQ
~~BUS_READY~~ BUS_READY
~~DATA_DRIVE~~ DATA_DRIVE
~~GND~~ GND
~~IRQ_REQ~~ IRQ_REQ
~~RP_BOOT_N~~ RP_BOOT_N
~~RP_DM~~ RP_DM
~~RP_DP~~ RP_DP
~~RP_RUN~~ RP_RUN
~~Z_A1~~ Z_A1
~~Z_A10~~ Z_A10
~~Z_A11~~ Z_A11
~~Z_A12~~ Z_A12
~~Z_A13~~ Z_A13
~~Z_A14~~ Z_A14
~~Z_A15~~ Z_A15
~~Z_A16~~ Z_A16
~~Z_A17~~ Z_A17
~~Z_A18~~ Z_A18
~~Z_A19~~ Z_A19
~~Z_A2~~ Z_A2
~~Z_A20~~ Z_A20
~~Z_A21~~ Z_A21
~~Z_A22~~ Z_A22
~~Z_A23~~ Z_A23
~~Z_A3~~ Z_A3
~~Z_A4~~ Z_A4
~~Z_A5~~ Z_A5
~~Z_A6~~ Z_A6
~~Z_A7~~ Z_A7
~~Z_A8~~ Z_A8
~~Z_A9~~ Z_A9
~~Z_AS_N~~ Z_AS_N
~~Z_CFGIN_N~~ Z_CFGIN_N
~~Z_D0~~ Z_D0
~~Z_D1~~ Z_D1
~~Z_D10~~ Z_D10
~~Z_D11~~ Z_D11
~~Z_D12~~ Z_D12
~~Z_D13~~ Z_D13
~~Z_D14~~ Z_D14
~~Z_D15~~ Z_D15
~~Z_D2~~ Z_D2
~~Z_D3~~ Z_D3
~~Z_D4~~ Z_D4
~~Z_D5~~ Z_D5
~~Z_D6~~ Z_D6
~~Z_D7~~ Z_D7
~~Z_D8~~ Z_D8
~~Z_D9~~ Z_D9
~~Z_LDS_N~~ Z_LDS_N
~~Z_READ~~ Z_READ
~~Z_UDS_N~~ Z_UDS_N

Core regulator placement MUST follow Raspberry Pi reference, including marked inductor orientation.

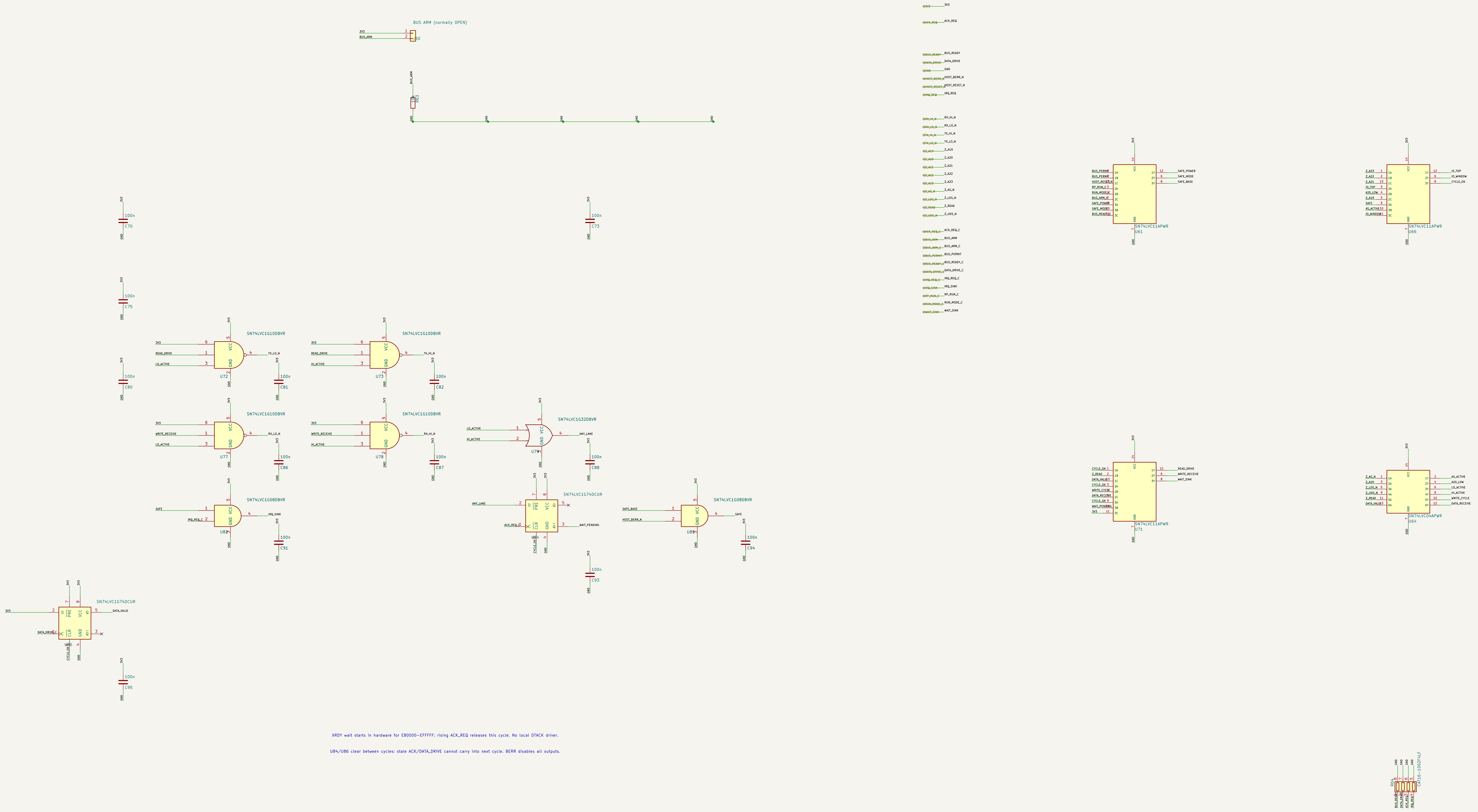
C26–C36: IOVDD/ADC/USB/QSPI decoupling. C37–C39: DVDD. Place at corresponding package pads.

FujiNet NIO
Sheet: /RP2350B and support/
File: rp.kicad_sch

Title: RP2350B and support

Size: A2
Date: 2026-11-09
Rev: R4 candidate
KiCad E.D.A. 10.0.7
Id: 9/10

Host monitor: 0.405V x 11 = 4.455V nominal trip; CT open = 20ms release delay. No host supply drawn for operation.


BUS ARM (normally OPEN)
3V3
BUS_ARM
J60
SN74LVC1G11DBVR
SN74LVC1G08DBVR
SN74LVC1G32DBVR
SN74LVC1G74DCUR
SN74LVC11APWR
SN74LVC04APWR
U61
U66
U71
U64
U72
U73
U77
U78
U79
U83
U89
U86
C70
C73
C75
C80
C81
C82
C86
C87
C88
C91
C93
C94
C95
100n
GND
READ_DRIVE
WRITE_RECEIVE
LO_ACTIVE
HI_ACTIVE
TX_LO_N
TX_HI_N
RX_LO_N
RX_HI_N
ANY_LANE
SAFE
IRQ_SINK
IRQ_REQ_C
ACK_REQ_C
WAIT_PENDING
SAFE_BASE
HOST_BERR_N
DATA_VALID
DATA_DRIVE
CYCLE_OK
SAFE_POWER
SAFE_MODE
IO_TOP
IO_WINDOW
A20_LOW
A0_ACTIVE

XRDY wait starts in hardware for E80000–EFFFFF; rising ACK_REQ releases this cycle. No local DTACK driver.

U84/U86 clear between cycles: stale ACK/DATA_DRIVE cannot carry into next cycle. BERR disables all outputs.

FujiNet NIO
Sheet: /Hardware bus interlocks/
File: control.kicad_sch
Title: Hardware bus interlocks
Size: A1
Date: 2026–11–09
Rev: R4 candidate
KiCad E.D.A. 10.0.7
Id: 10/10